

Configure **VLANs** on the FortiGate firewall, **assign an IP address** to each VLAN, and enable the **DHCP** server to provide automatic IP assignment to devices within each VLAN.

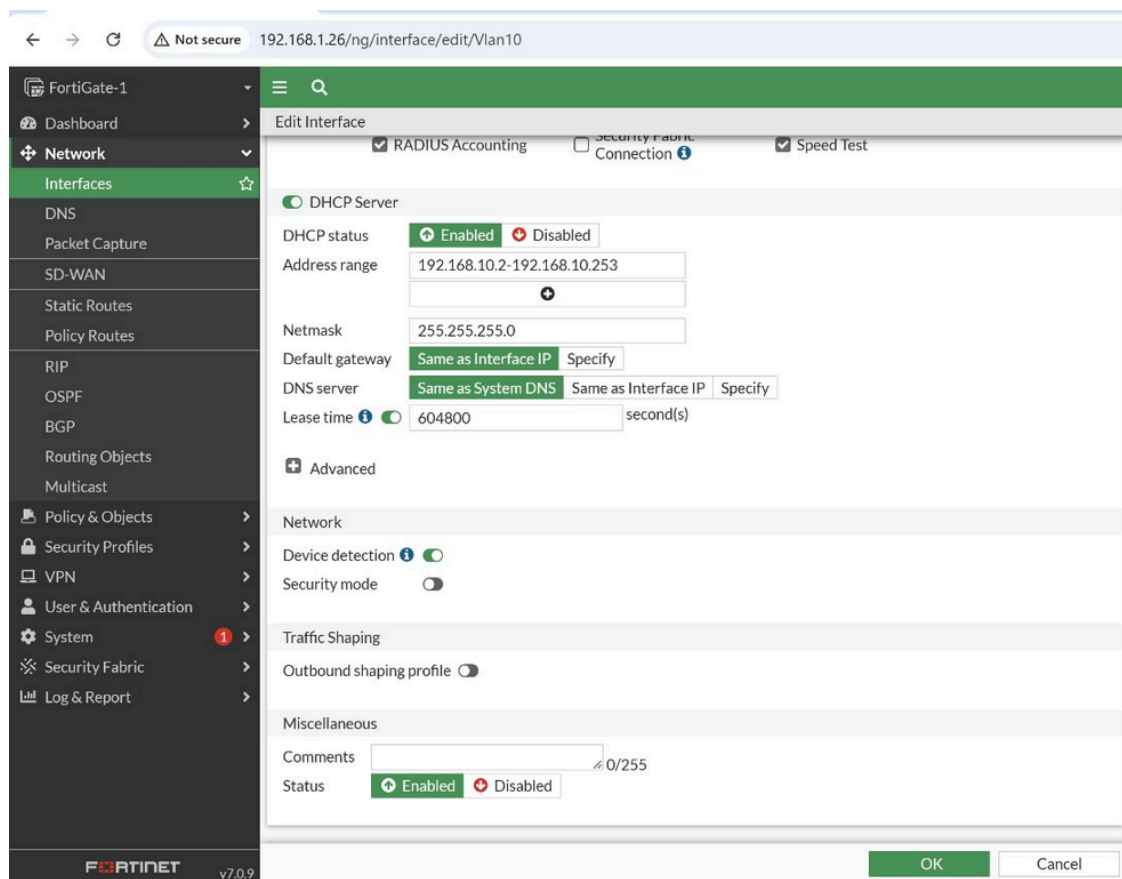
>>>Network >>> Interfaces >>> Create new

Vlans for HQ Fortigate

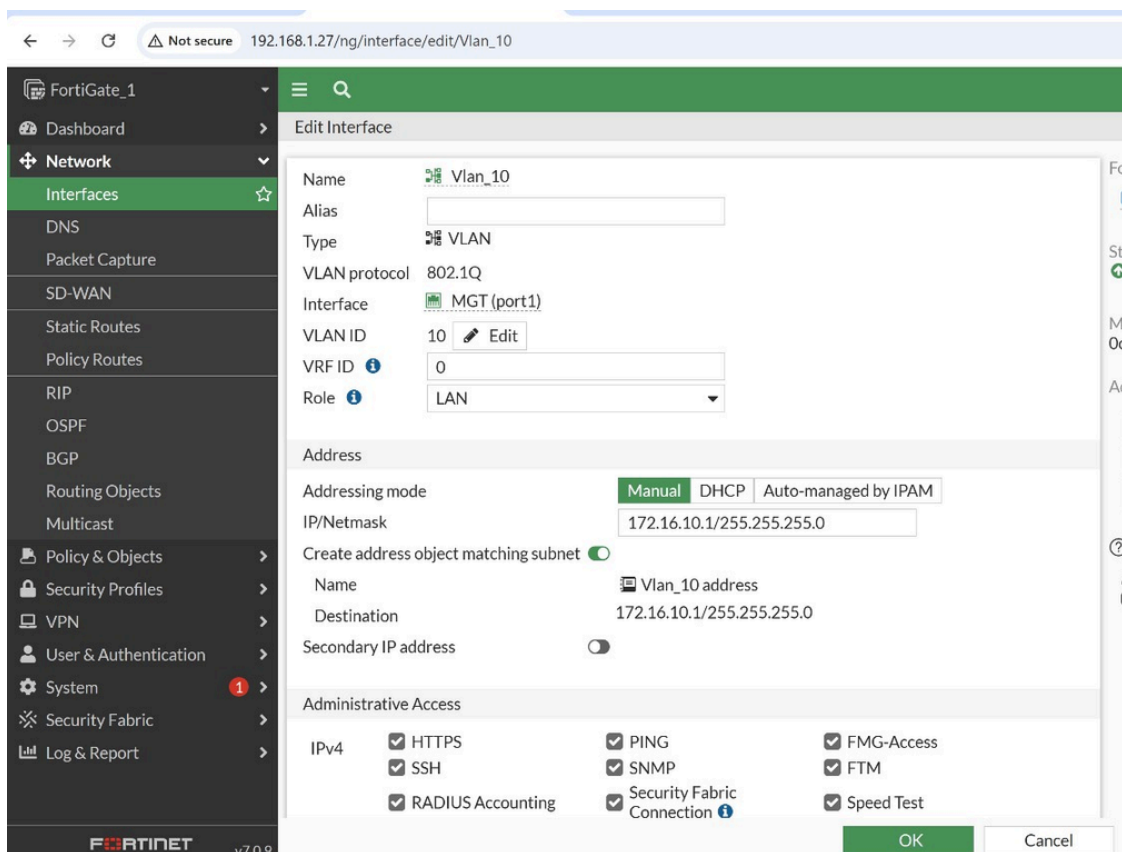
The screenshot displays the FortiGate web interface for editing a VLAN interface. The left sidebar shows the navigation menu with 'Network' > 'Interfaces' selected. The main panel is titled 'Edit Interface' and shows the configuration for 'Vlan10'. The 'Name' is 'Vlan10', 'Type' is 'VLAN', 'VLAN protocol' is '802.1Q', 'Interface' is 'port1', 'VLAN ID' is '10', 'VRF ID' is '0', and 'Role' is 'LAN'. The 'Address' section shows 'Manual' addressing mode with IP '192.168.10.1/255.255.255.0'. The 'DHCP Server' is enabled. The 'Administrative Access' section shows various services enabled, including 'HTTPS', 'SSH', 'RADIUS Accounting', 'PING', 'SNMP', 'Security Fabric Connection', 'FMG-Access', 'FTM', and 'Speed Test'.

Section	Field	Value
General	Name	Vlan10
	Alias	
	Type	VLAN
	VLAN protocol	802.1Q
	Interface	port1
	VLAN ID	10
VRF	VRF ID	0
	Role	LAN
	Address	
	Addressing mode: Manual (selected), DHCP, Auto-managed by IPAM	
IP Address	IP/Netmask	192.168.10.1/255.255.255.0
	Create address object matching subnet	☒
	Name	Vlan10 address
	Destination	192.168.10.1/255.255.255.0
Secondary IP	Secondary IP address	☐
	Administrative Access	
Access	IPv4	☒ HTTPS
		☒ SSH
		☒ RADIUS Accounting
		☒ PING
		☒ SNMP
		☐ Security Fabric Connection
		☒ FMG-Access
		☒ FTM
		☒ Speed Test
	DHCP Server	
DHCP status: ☒ Enabled, ☐ Disabled		

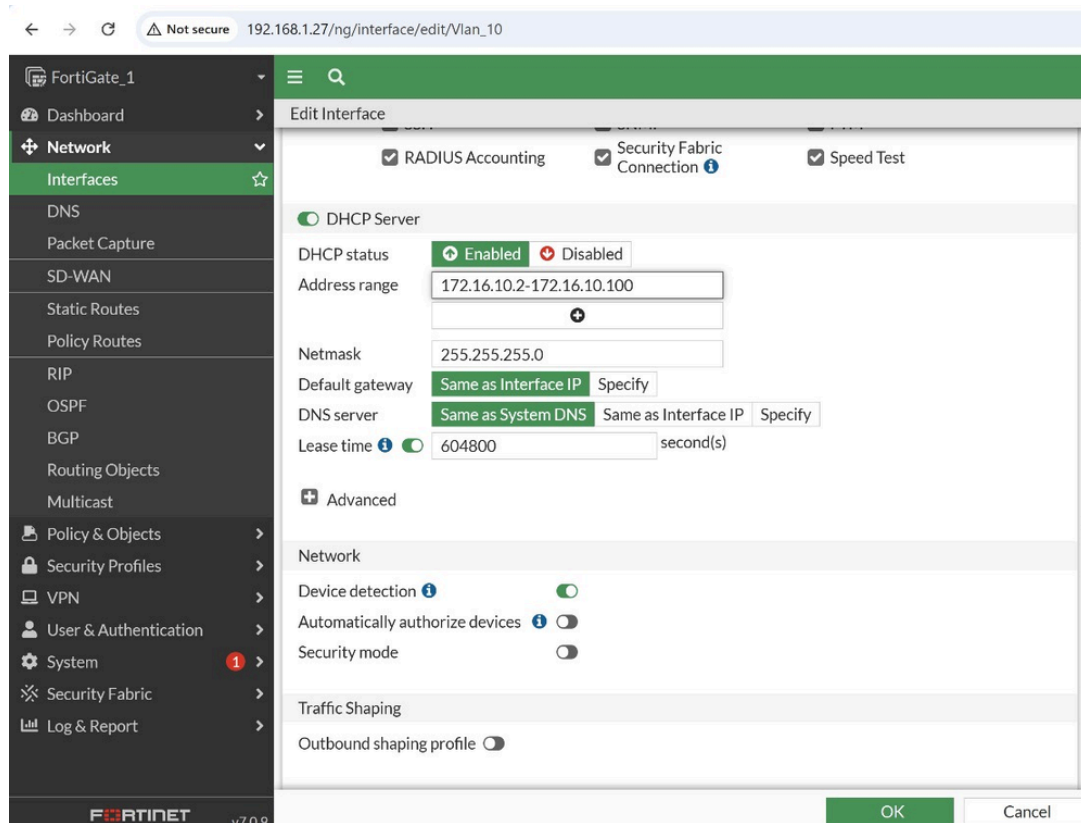
enable dhcp server



Vlans for BR Fortigate



enable dhcp server



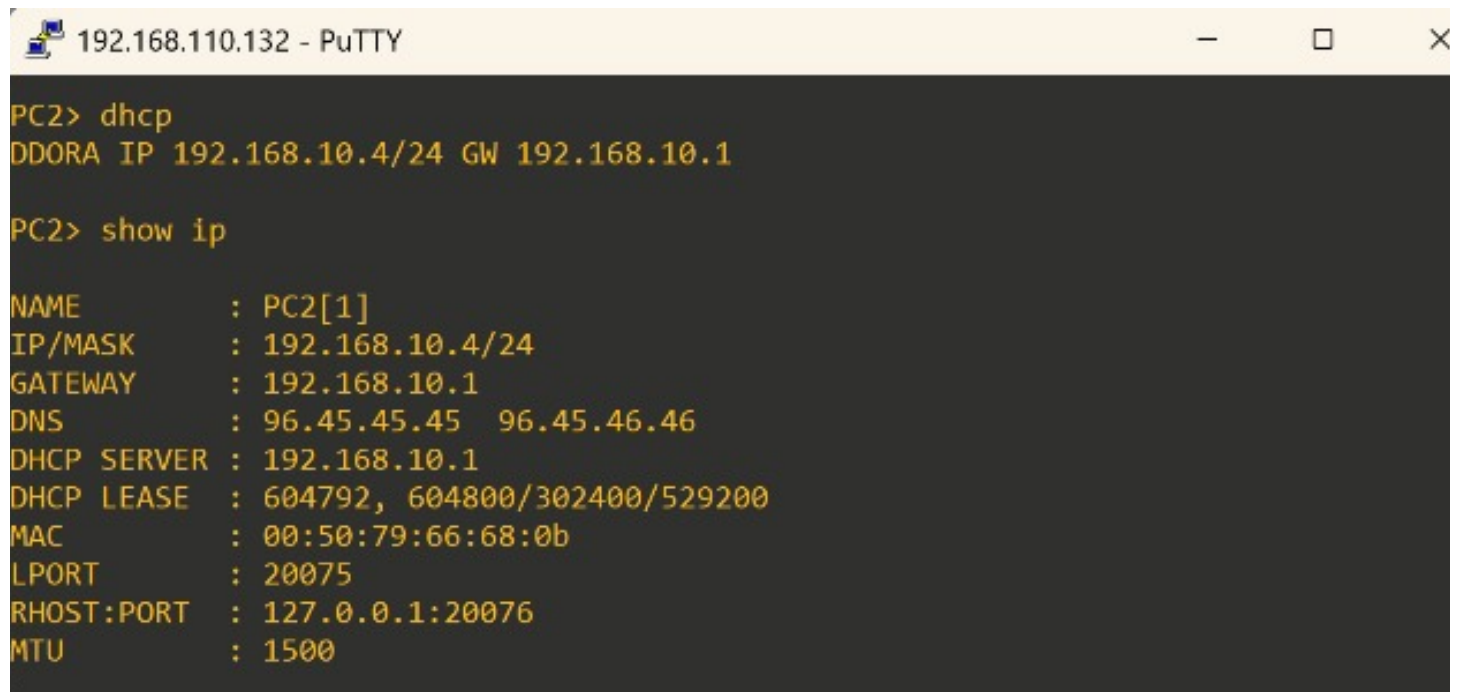
Assign ips of Vlans on Switch (HQ)

```
!
interface Vlan10
 ip address 192.168.10.254 255.255.255.0
 ip helper-address 192.168.10.1
!
interface Vlan20
 ip address 192.168.20.254 255.255.255.0
 ip helper-address 192.168.20.1
!
interface Vlan30
 ip address 192.168.30.254 255.255.255.0
 ip helper-address 192.168.30.1
!
interface Vlan40
 ip address 192.168.40.254 255.255.255.0
 ip helper-address 192.168.40.1
!
interface Vlan50
 ip address 192.168.50.254 255.255.255.0
 ip helper-address 192.168.50.1
!
interface Vlan60
 ip address 192.168.60.254 255.255.255.0
 ip helper-address 192.168.60.1
!
```

Assign ips of Vlans on Switch (BR)

```
interface Vlan10
 ip address 172.16.10.254 255.255.255.0
 ip helper-address 172.16.10.1
!
interface Vlan20
 ip address 172.16.20.254 255.255.255.0
 ip helper-address 172.16.20.1
!
interface Vlan30
 ip address 172.16.30.254 255.255.255.0
 ip helper-address 172.16.30.1
!
interface Vlan40
 ip address 172.16.40.254 255.255.255.0
 ip helper-address 172.16.40.1
!
interface Vlan50
 ip address 172.16.50.254 255.255.255.0
 ip helper-address 172.16.50.1
!
interface Vlan60
 ip address 172.16.60.254 255.255.255.0
 ip helper-address 172.16.60.1
!
ip forward protocol nd
```

Test PCs on (HQ)



The screenshot shows a PuTTY terminal window titled "192.168.110.132 - PuTTY". The terminal displays the following commands and output:

```
PC2> dhcp
DDORA IP 192.168.10.4/24 GW 192.168.10.1

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.168.10.4/24
GATEWAY    : 192.168.10.1
DNS        : 96.45.45.45 96.45.46.46
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 604792, 604800/302400/529200
MAC        : 00:50:79:66:68:0b
LPORT      : 20075
RHOST:PORT  : 127.0.0.1:20076
MTU        : 1500
```

Test PCs on (BR)

```
192.168.110.132 - PuTTY

Press '?' to get help.

Executing the startup file

PC13> dhcp
DDORA IP 172.16.10.2/24 GW 172.16.10.1

PC13> show ip

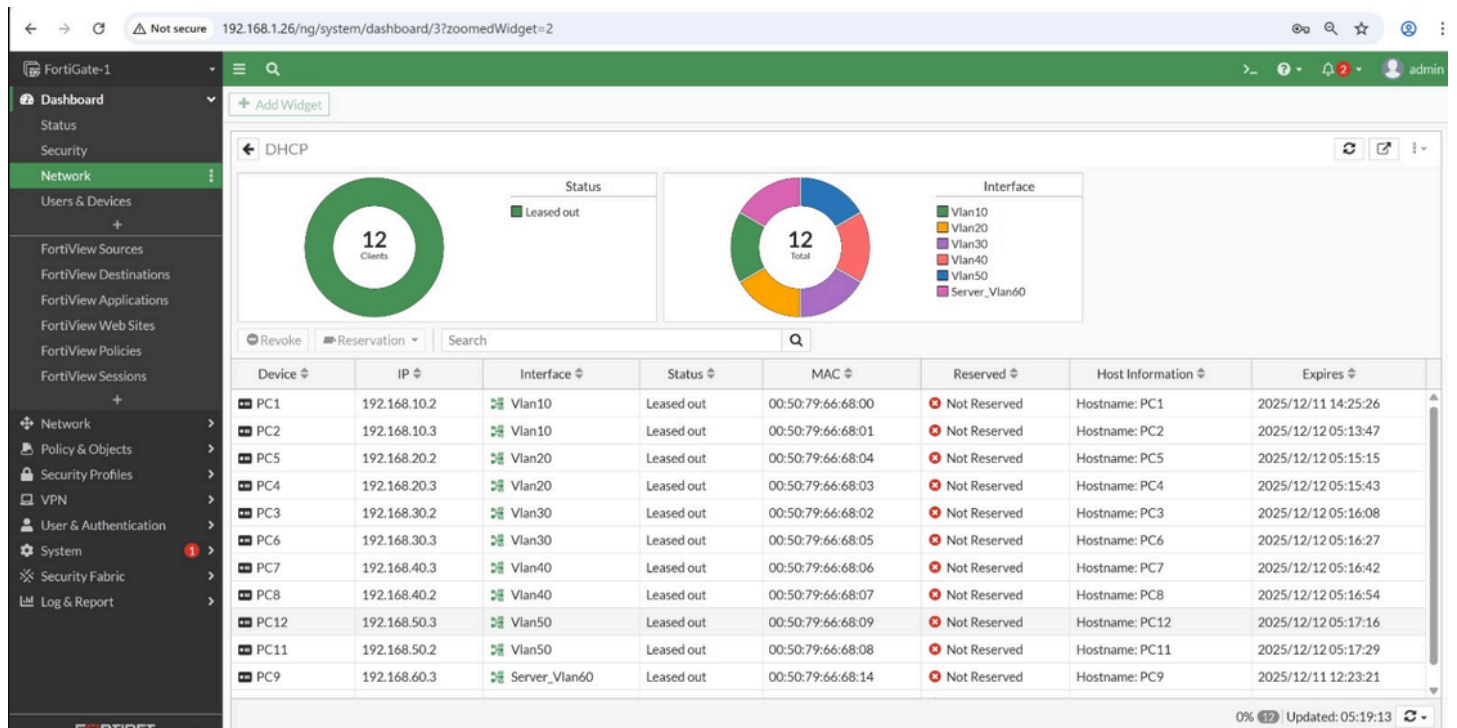
NAME          : PC13[1]
IP/MASK       : 172.16.10.2/24
GATEWAY       : 172.16.10.1
DNS           : 96.45.45.45 96.45.46.46
DHCP SERVER   : 172.16.10.1
DHCP LEASE    : 604795, 604800/302400/529200
MAC           : 00:50:79:66:68:0a
LPORT        : 20132
RHOST:PORT    : 127.0.0.1:20133
MTU           : 1500
```

On HQ >>> Vlan 60 dhcp server
range ips **[192.168.60.2 - 192.168.60.254]** excluded
192.168.60.10 (static ip of server)

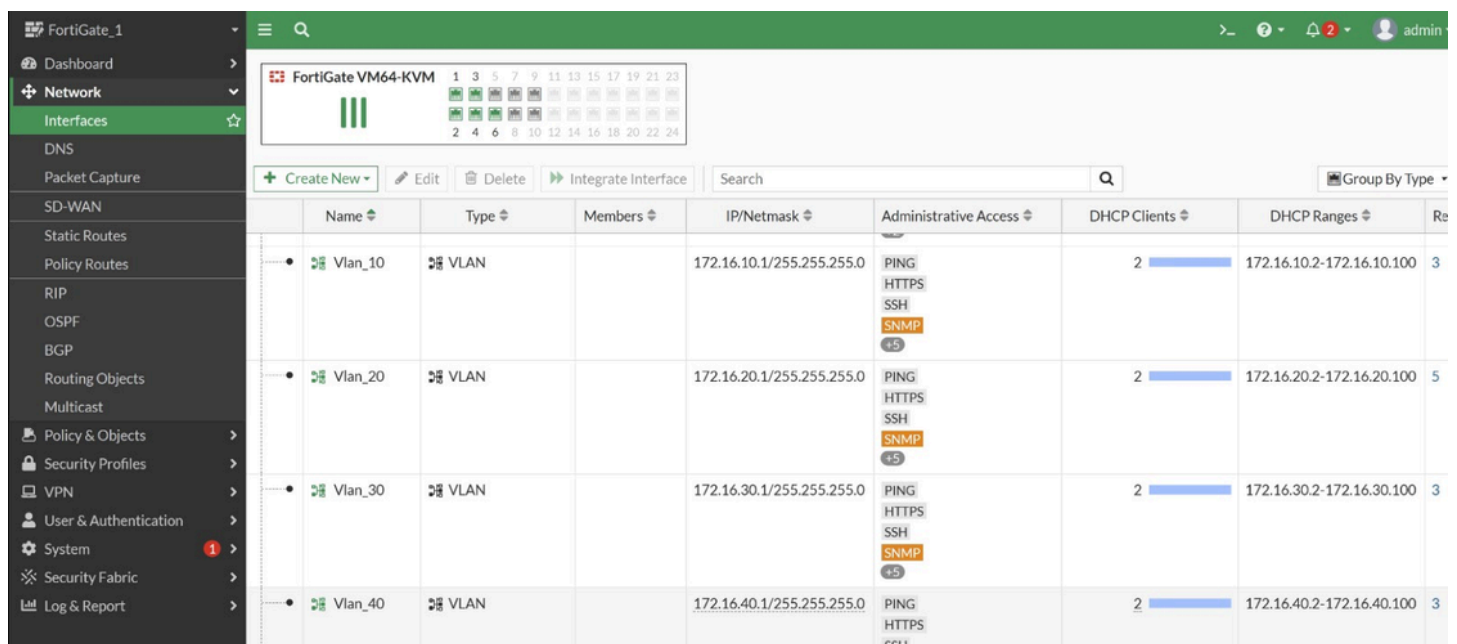
The screenshot shows the FortiGate VM64-KVM web interface. The left sidebar contains a navigation menu with categories like Network, Policy & Objects, Security Profiles, VPN, User & Authentication, System, Security Fabric, and Log & Report. The 'Network' section is expanded, showing 'Interfaces' as the selected option. The main content area displays a table of interfaces. The table has columns for Name, Type, Members, IP/Netmask, Administrative Access, DHCP Clients, DHCP Ranges, and Ref. The 'Server_Vlan60' interface is highlighted, showing it is a VLAN type with IP/Netmask 192.168.60.1/255.255.255.0. The 'Administrative Access' column lists PING, HTTPS, SSH, and SNMP. The 'DHCP Clients' column shows 3 clients. The 'DHCP Ranges' column shows the range 192.168.60.2-192.168.60.9 and 192.168.60.11-192.168.60.254. The 'Ref.' column shows 5.

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
port1	Physical Interface		0.0.0.0/0.0.0.0	PING HTTPS SSH SNMP			12
Server_Vlan60	VLAN		192.168.60.1/255.255.255.0	PING HTTPS SSH SNMP	3	192.168.60.2-192.168.60.9 192.168.60.11-192.168.60.254	5
Vlan10	VLAN		192.168.10.1/255.255.255.0	PING HTTPS SSH SNMP	3	192.168.10.2-192.168.10.253	3
Vlan20	VLAN		192.168.20.1/255.255.255.0	PING HTTPS SSH SNMP	3	192.168.20.2-192.168.20.253	3
Vlan30	VLAN		192.168.30.1/255.255.255.0	PING	2	192.168.30.2-192.168.30.253	2

>>> Dashboard >>> Network >>> DHCP



On BR



>>> Dashboard >>> Network >>> DHCP

FortiGate_1

Dashboard

Status

Security

Network

Users & Devices

FortiView Sources

FortiView Destinations

FortiView Applications

FortiView Web Sites

FortiView Policies

FortiView Sessions

Network

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

192.168.1.27/ng/system/dashboard/3?zoomedWidget=2

admin

+

+

DHCP

12 Clients

Leased out

12 Total

Vlan_10

Vlan_20

Vlan_30

Vlan_50

Vlan_60

Revoke

Reservation

Search

Device	IP	Interface	Status	MAC	Reserved	Host Information	Expires
PC13	172.16.10.2	Vlan_10	Leased out	00:50:79:66:68:0a	Not Reserved	Hostname: PC13	2025/12/12 04:43:53
PC14	172.16.10.3	Vlan_10	Leased out	00:50:79:66:68:0b	Not Reserved	Hostname: PC14	2025/12/12 04:44:45
PC16	172.16.20.3	Vlan_20	Leased out	00:50:79:66:68:0d	Not Reserved	Hostname: PC16	2025/12/11 12:24:16
PC15	172.16.20.2	Vlan_20	Leased out	00:50:79:66:68:0c	Not Reserved	Hostname: PC15	2025/12/11 12:39:34
PC17	172.16.30.2	Vlan_30	Leased out	00:50:79:66:68:0e	Not Reserved	Hostname: PC17	2025/12/12 04:45:00
PC18	172.16.30.3	Vlan_30	Leased out	00:50:79:66:68:0f	Not Reserved	Hostname: PC18	2025/12/12 04:45:19
PC21	172.16.50.2	Vlan_50	Leased out	00:50:79:66:68:12	Not Reserved	Hostname: PC21	2025/12/12 04:47:03
PC22	172.16.50.3	Vlan_50	Leased out	00:50:79:66:68:13	Not Reserved	Hostname: PC22	2025/12/12 05:03:31

0% Updated: 05:09:50